Q1. Big Data refers to very large and complex data sets that cannot be processed using traditional data processing tools. It requires advanced technologies like Hadoop for storage and analysis.

Q2. MapReduce is a programming model in Hadoop used to process large data sets. It divides the task into two phases: Map (processing data) and Reduce (combining results).

Q3. The 5 V's of Big Data are Volume (size of data), Velocity (speed of data), Variety (different types), Veracity (accuracy), and Value (usefulness).

Q4. The types of Big Data are Structured (organized data), Unstructured (raw data like images, videos), and Semi-structured (partially organized data like JSON).

Q5. Hadoop is an open-source framework used to store and process large data sets across distributed systems using HDFS and MapReduce.

Q6. Think of HDFS like a **big library system** 📚

- One **main librarian (NameNode)** → knows where every book is
- Many **storage rooms (DataNodes)** → store the actual books

Components of HDFS Architecture

1. NameNode (Master)

- Manages the entire system
- Stores metadata (file names, locations)
- Does NOT store actual data

Example: Like a **manager telling where files are stored**

2. DataNode (Workers)

- Store the actual data (blocks)
- Perform read/write operations
- Multiple DataNodes in cluster

Example: Like **storage workers holding data**

3. Secondary NameNode

- Takes backup of NameNode metadata
- Helps in recovery

Not a replacement, just **support system**

How HDFS Works (Flow)

1. File is divided into **small blocks (128MB)**
2. Blocks are stored in different DataNodes
3. NameNode keeps track of block locations
4. Client requests file → NameNode gives location → DataNode sends data

Key Features

- Distributed storage
- Fault tolerance
- High scalability

Q7. Replication = **Making copies of data**

Like saving your file in **Google Drive + Phone + Laptop**

Why Replication is Important

1. **Fault Tolerance**
    - If one DataNode fails → data still available
2. **High Availability**
    - Data can be accessed anytime
3. **Reliability**
    - Prevents data loss
4. **Better Performance**
    - Data can be read from nearest copy

How HDFS Ensures Data Availability

1. Replication Factor

- Default = **3 copies**
- Stored on different DataNodes

2. Heartbeat Mechanism

- DataNodes send signals to NameNode
- If no signal → node is considered failed

3. Automatic Re-replication

- If a DataNode fails:
    - NameNode creates new copies on other nodes

4. Rack Awareness

- Copies stored in different racks
- Protects from network/rack failure

5. Data Recovery Process

1. DataNode fails
2. NameNode detects failure
3. New copies created automatically
4. System continues without interruption

Conclusion (Write this for full marks)

HDFS uses replication and automatic recovery mechanisms to ensure data availability, fault tolerance, and reliability even when nodes fail.